The Balance -Volume Trade-off: Is Hierarchical K-Means Sampling Effective for Constrained Data Budgets

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Abstract

Dataset imbalance remains a critical bottleneck in computer vision, particularly when fine-tuning or training lightweight models on restricted data budgets. While hierarchical k-means sampling has proven effective for flattening class distributions on massive datasets, its viability for smaller-scale applications remains underexplored. In this paper, we investigate the efficacy of this sampling technique on constrained, long-tailed vision datasets. We systematically evaluate the impact of embedding representations on clustering quality, analyze performance trade-offs across varying data budgets, and benchmark the method for models trained from scratch. We conclude that sacrificing overall data volume for class balance is not universally beneficial; rather, it is a valid trade-off exclusively for severely skewed distributions. Furthermore, we demonstrate that standard supervised embeddings (ResNet) outperform self-supervised representations (DINOv2) as a basis for clustering-driven sampling, generalizing more robustly across diverse architectures.

1. Introduction

Long-tailed distributions continue to dominate domain-specific fine-tuning and resource-constrained model training from scratch. While recent work addresses this via hierarchical k-means sampling, it relies heavily on massive datasets and extensive compute budgets. We investigate this sampling technique on smaller datasets to evaluate the balance-volume trade-off and analyze the impact of different embedding representations.

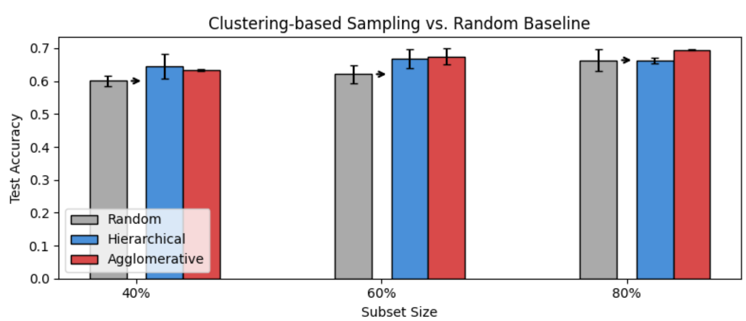

Figure 1: Comparison of clustering-based sampling strategies against random baseline (Experiment 3). Agglomerative clustering emerges as a competitive alternative to hierarchical k-means, achieving the highest accuracy at most tested subset sizes.

2. Related Works

Recent advances in dataset curation increasingly rely on embedding spaces to identify optimal data subsets and mitigate the computational costs of training [1]. Within this domain, core-set selection and active learning methods frequently utilize geometric or clustering-based heuristics to preserve semantic diversity [2]. Most notably, a hierarchical k-means approach was introduced for automatic data curation in self-supervised learning [3], demonstrating remarkable efficacy in balancing the massive datasets required to train large foundational architectures like DINOv3 [4]. However, the robustness of such semantic-aware partitioning when operating under strictly constrained data budgets and varying feature representations remains largely underexplored.

3. Methodology

Our approach mitigates long-tailed dataset distributions by performing concept-aware data sampling. The framework consists of three sequential stages: mapping raw images to a high-dimensional feature space using a choice of embeddings, grouping semantically similar instances via recursive hierarchical clustering, and applying a balancing sampling strategy to extract a uniform training subset. The sampling method is described in more detail in [3].

4. Experiments

4.1 Experimental Setup

We evaluate ResNet-18 and ResNet-50 via fine-tuning, training from scratch, and linear probing. We sampled 10-class subsets from ImageNet-1K (13,000 images) and Places365 ($\sim$ 20,000 images) using a Dirichlet distribution ($\alpha \in \{0.5, 1.0, 2.0\}$). To prevent class disappearance, we enforced a 200-image minimum threshold. Validation and testing sets (Imagenette and Places365 splits) were kept strictly balanced for global accuracy evaluation. Clustering hyperparameters were optimized via Optuna.

We conducted three primary experiments:

1. **Embeddings Comparison:** We investigate how feature embeddings affect sampling accuracy by evaluating two distinct extractors: ResNet-50 and DINOv2-Large. The selected subsets train a ResNet-18 model evaluated on uniform Imagenette (averaged over three random seeds against a random sampling baseline) and heavily skewed ImageNet-LT (single seed evaluation).

2. **Varying Data Budgets:** To assess the method's performance across varying data budgets, we measured ResNet-50 performance benchmarked on the skewed ImageNet-1K dataset sampled from the Dirichlet distribution. The hierarchical k-means sampling was constrained to retain from 50% to 90% of the original data volume. We conducted the experiment across five random seeds.

   The ResNet-50 model was evaluated using both fine-tuning and linear probing strategies. For the hierarchical k-means sampling process in this experiment, ResNet-50 embeddings were utilized.

3. **Training from Scratch:** To assess the impact of class balancing prior to training from scratch, we conducted experiments using the skewed variant of the Places365 dataset ($\alpha = 0.5$). We used the ResNet-18 architecture (not ResNet-50), as the restricted data renders higher-capacity models susceptible to overfitting. DINOv2 embeddings were employed for feature extraction. We benchmarked result on data from hierarchical k-means sampling against one generated using agglomerative and regular k-means clustering. We also tested data form sampled randomly as a baseline.

4.2 Experimental Results

Embeddings & Data Budgets: On Imagenette, semantic-aware sampling (DINO embeddings) peaks at 90% retention (98.70%±0.08), surpassing the 100% baseline but losing to random sampling. Conversely, on the skewed ImageNet-LT, standard ResNet features drastically outperform DINO representations across all budgets (e.g., 61.57% vs. 27.14% at a 40% budget), indicating supervised features better capture tail-class semantic boundaries.

Impact of Skewness: Figure 2 illustrates that aggressive data reduction (e.g., 50%) is a valid trade-off exclusively for extreme skew ($\alpha = 0.5$), yielding potential improvements but introducing high variance due to the reduced dataset size. For datasets with milder imbalances ($\alpha = 1.0, 2.0$), hierarchical k-means sampling failed to produce significant improvements. Similar results were shown on uniform dataset - Imagenette where random sampling was often better than hierarchical sampling.

Table 1: Classification accuracy comparison across different data selection percentages.

(a) Results on Imagenette

Subset (%)	DINO	ResNet	Random
10	86.10 ± 1.49	94.60 ± 0.70	**97.57 ± 0.09**
20	91.90 ± 0.75	97.63 ± 0.09	**98.03 ± 0.12**
30	93.50 ± 0.93	97.80 ± 0.14	**98.03 ± 0.05**
40	95.90 ± 0.51	**98.23 ± 0.05**	98.20 ± 0.08
50	97.37 ± 0.21	98.20 ± 0.08	**98.43 ± 0.05**
60	98.17 ± 0.12	98.50 ± 0.14	**98.53 ± 0.05**
70	98.37 ± 0.17	**98.40 ± 0.00**	98.40 ± 0.08
80	98.40 ± 0.14	**98.57 ± 0.09**	98.56 ± 0.05
90	98.70 ± 0.08	98.60 ± 0.00	**98.73 ± 0.05**
100	**98.63 ± 0.12**	**98.63 ± 0.12**	**98.63 ± 0.12**

(b) Results on ImageNet-LT

Subset (%)	DINO	ResNet
30	42.63	**48.24**
40	27.14	**61.57**
70	71.56	**74.28**
80	74.61	**76.03**
90	75.14	**75.86**
100	**79.38**	**79.38**

Bold values indicate the highest accuracy among the evaluated methods for a given subset. Underlined values denote results that surpass the baseline accuracy achieved when training on the entire 100% dataset.

Competitive Alternative: Table 2 shows that agglomerative sampling achieves the highest accuracy at three out of four subset sizes (20%, 60%, 80%), while hierarchical k-means surpasses it only at the 40% budget. Notably, agglomerative clustering at 80% retention (0.695 ± 0.002) exceeds the 100% baseline. Both clustering methods consistently outperform random sampling, confirming the value of semantic-aware data selection under constrained, long-tailed conditions.

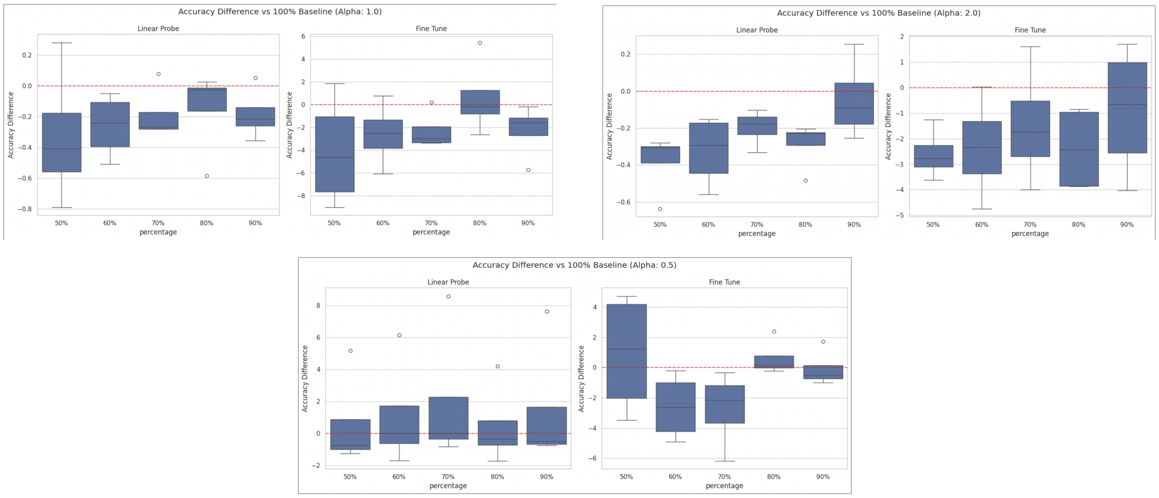

Figure 2: Accuracy difference vs. 100% baseline across varying data budgets. Top left: $\alpha = 1.0$. Top right: $\alpha = 2.0$. Bottom: The highly skewed $\alpha = 0.5$ distribution, demonstrating the most substantial variance and potential improvement.

Table 2: Classification accuracy of ResNet-18 trained from scratch on the Places365 subset ($\alpha = 0.5$) across different clustering-based sampling strategies.

Subset (%)	Agglomerative	Hierarchical	KMeans	Random
20	**0.630 ± 0.017**	0.604 ± 0.037	0.529 ± 0.033	0.548 ± 0.013
40	0.634 ± 0.003	**0.645 ± 0.036**	0.608 ± 0.014	0.601 ± 0.016
60	**0.674 ± 0.024**	0.669 ± 0.029	0.669 ± 0.002	0.621 ± 0.028
80	**0.695 ± 0.002**	0.662 ± 0.008	0.680 ± 0.000	0.664 ± 0.033
100		0.685 ± 0.004		

Bold values indicate the highest accuracy among the evaluated methods for a given subset. Underlined values denote results that surpass the baseline accuracy achieved when training on the entire 100% dataset.

5. Conclusions

In this paper, we evaluated hierarchical k-means sampling under constrained data budgets, demonstrating that sacrificing dataset volume for class balance yields performance gains exclusively in severely skewed distributions ($\alpha = 0.5$). Furthermore, our experiments reveal that supervised feature spaces capture class boundaries more effectively than self-supervised representations in low-budget regimes, while agglomerative clustering emerges as a powerful alternative. However, a primary limitation of this study is the restricted computational and time budget, which confined our empirical evaluation to smaller-scale subsets and fewer hyperparameter configurations. Future work can focus on scaling these experiments to larger datasets and broader model architectures to thoroughly map out the boundary conditions of this data curation technique.

References

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